

Hoang (Bolton) Tran, Ph.D.

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Education

Pennsylvania State University , Ph.D. in Chemical Engineering Thesis: Electrocatalysis at the Electrode/Electrolyte Interface: a Multiscale Molecular Model Advisors: Prof. Michael J. Janik & Prof. Scott T. Milner	Sept 2018 – Aug 2023
Drexel University , B.S. in Chemical Engineering	Sept 2012 – June 2017

Research

Postdoctoral research fellow University of Michigan, Department of Chemical Engineering Advisor: Prof. Bryan R. Goldsmith	Sept 2023 – present
Graduate research assistant Pennsylvania State University, Department of Chemical Engineering Advisors: Prof. Michael J. Janik & Prof. Scott T. Milner	Sept 2018 – Aug 2023
Undergraduate research assistant Drexel University, Department of Chemical Engineering Advisors: Prof. Jason Baxter	Sept 2014 – Apr 2015

Industry

Avid Radiopharmaceuticals , Philadelphia, PA Quality assurance associate Manufacturing intern	Nov 2016 – Aug 2018 Mar 2015 – Aug 2016
Zeolyst International , Conshohocken, PA Analytical intern	May 2014 – Sept 2014

Teaching

Atomistic scale simulations , ME/CHE 505, Pennsylvania State University Guest lecturer (six one-hour lectures)	Spring 2022
Simulation techniques and applications , CHE 597, Pennsylvania State University Guest lecturer (a one-hour lecture)	Fall 2022
Material balance , CHE 210, Pennsylvania State University Graduate teaching assistant	Fall 2022
Reaction engineering , CHE 430, Pennsylvania State University Graduate teaching assistant	Spring 2021

Research Mentoring

Pennsylvania State University

- Elizabeth Long, Women In Science and Engineering Research (WISER) undergraduate researcher
- Yusheng Cai, undergraduate researcher, now a Ph.D. candidate at the University of Pennsylvania
- Joe Hughes, undergraduate researcher, now an engineer at Naval Reactors

University of Michigan

- Ankit Mathanker, 4th year Ph.D. candidate
- Dean Sweeney, 2nd year Ph.D. candidate
- Roshini Dantuluri, 2nd year Ph.D. candidate
- Yifei Liu, undergraduate researcher
- Mad Lindsey, undergraduate researcher
- Jean-Patrick Selo, community college researcher

Publications

Published peer-reviewed papers

1. **Tran, B.**, Janik, M. J., Milner, S. T., “Hydration-shell Solvation and Screening Govern Alkali Cation Concentrations at Electrochemical Interfaces”, *J. Phys. Chem. C*, <https://doi.org/10.1021/acs.jpcc.4c05537>.
2. Manthanker, A., Halarnkar, S., **Tran, B.**, Signh, N., Goldsmith, B. R., “Synergistic Effects in Organic Mixtures for Enhanced Catalytic Hydrogenation and Hydrodeoxygenation”, *Chem Catalysis* 101135, Online 10/16/24, <https://doi.org/10.1016/j.checat.2024.101135>.
3. **Tran, B.**, Goldsmith, B. R., “Theoretical Investigation of the Potential-Dependent CO Adsorption on Copper Electrodes”, *J. Phys. Chem. Lett.* **15**, 6538–6543 (2024)
4. Wong, A. J., **Tran, B.**, Agrawal, N., Goldsmith, B. R., Janik, M. J., “Sensitivity Analysis of Electrochemical Double Layer Approximations on Electrokinetic Predictions: Case Study for CO Reduction on Copper”, *J. Phys. Chem. C* **128**, 10837–10847 (2024)
5. **Tran, B.**, Zhou, Y., Janik, M. J., Milner, S. T., “Negative Dielectric Constant of Water at a Metal Interface”, *Phys. Rev. Lett.* **131**, 248001 (2023)
6. Ostervold, L., Daneshpour, R., Facchinei, M., **Tran, B.**, Wetherington, M., Alexopoulos, K., Greenlee, L., Janik, M. J., “Identifying the Local Atomic Environment of the “Cu³⁺” State in Alkaline Electrochemical Systems”, *ACS Appl. Mater. Interfaces* **15**, 27878–27892 (2023)
7. **Tran, B.**, Milner, S. T., Janik, M. J., “Kinetics of Acid-Catalyzed Dehydration of Alcohols in Mixed Solvent Modeled by Multiscale DFT/MD”, *ACS Catal.* **12**, 13193–13206 (2022)
8. **Tran, B.**, Cai, Y., Janik, M. J., Milner, S. T., “Hydrogen Bond Thermodynamics in Aqueous Acid Solutions: A Combined DFT and Classical Force-Field Approach”, *J. Phys. Chem. A* **126**, 7382–7398 (2022)
9. Edley, M. E., Opananont, B., Conley, J. T., **Tran, H.**, Smolin, S. Y., Li, S., Dillon, A. D., Fafarman, A. T., Baxter, J. B., “Solution Processed CuSbS₂ Films for Solar Cell Applications”, *Thin Solid Films* **646**, 180–189 (2018)

Manuscripts in revision, submission, or preparation

1. Long, E., **Tran, B.**, Milner, S. T., “Tuning Partial Charges of Alkyl Alcohols to Improve Simulated Fluid Properties”, *J. Chem. Phys.*, manuscript in revision.
2. Wong, A. J., **Tran, B.**, Milner, S. T., Janik, M. J., “Electrode-Electrolyte Interfacial Effects on Specific Alkali Metal Cation Adsorption to Late Transition Metal Surfaces”, manuscript in preparation.
3. Manthanker, A., Sharma, G., **Tran, B.**, Signh, N., Goldsmith, B. R., “Effect of Ions on the Aqueous-Phase Adsorption of Benzene, Phenol, and Catechol on Ag(111)”, manuscript in preparation.
4. **Tran, B.**, Goldsmith, B. R., “Predicting Competitive Anion Electrosorption on Late Transition Metal Surfaces”, manuscript in preparation.
5. **Tran, B.**, Sweeney, D., Selo, J., Liu, Y., Lindsey, M., Goldsmith, B. R., “Effects of hydration on the predictive power of O-H stretching frequency on acid pKa”, manuscript in preparation.

Conference Presentation

1. **AIChE 2024** | San Diego | oral & poster presentations.
2. **ACS Fall 2024** | Denver | oral & poster presentations.
3. **GRC/GRS 2024** | New London | oral (invited) & poster presentations.
4. **Michigan Catalysis Society 2024** | Ann Arbor | oral presentation.
5. **ACS Spring 2023** | Indianapolis | oral presentation.
6. **AIChE 2022** | Phoenix | oral & poster presentations.
7. **Pittsburgh-Cleveland Catalysis Society 2022** | State College | poster presentation.
8. **North American Catalysis Society 2022** | New York | oral & poster presentations.
9. **ACS Spring 2022** | Virtual | oral presentation.
10. **APS March 2022** | Chicago | oral presentation.
11. **AIChE 2021** | Boston | oral presentation.

Awards

- University of Michigan postdoctoral travel award | ACS Fall 2024
- ACS CATL ChemCatBio travel award | ACS Spring 2023
- Poster presentation 1st place | Pittsburg-Cleveland Catalysis Society 2022
- FGSA travel award | APS March 2022
- Departmental best qualifying exam | Pennsylvania State University 2019

Broader Impact Activities

- Volunteered at the Midwest Thermodynamics and Statistical Mechanics Conference in Ann Arbor, 2024.
- Guest-spoke at the University of Michigan Center for Simulation and Data Intensive Research in Chemical Engineering, 2024.
- Poster-judged for University of Michigan Graduate Research Symposium 2023.
- Founded and led a Vietnamese Graduate Student Association at Pennsylvania State University.
- Assisted newly admitted ChE graduate students in Pennsylvania State University departmental mentorship program.